

# Runbook — Confluent Cloud for Apache Flink (CCAF)

End-to-end operational guide for running the `confluent-kafka-isotope` reports on **Confluent Cloud for Apache Flink (CCAF)**: provision → deploy reports → drive traffic → observe → teardown. Unlike the [Confluent Platform + Flink on minikube](#), this is **Terraform-driven** — no local cluster — and everything runs in a fresh Confluent Cloud environment under `terraform/`.

This is the **managed CCAF** path. The self-managed **Confluent Platform + Flink on minikube** path is in [docs/runbook-minikube.md](#). Both runtimes run the same seven reports from the same PTF JAR — see [root README §3.3](#) for the format-by-runtime split.

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## 1.0 Prerequisites

- `Terraform` `>= 1.13` installed locally.
- A **Cloud-level** Confluent Cloud API key (not cluster-scoped) with permission to create environments, Kafka clusters, Flink compute pools, service accounts, role bindings, Flink artifacts, and statements. Generate via Console → Settings → Cloud API keys.

## 2.0 Provision + deploy the reports

```
make cc-flink-reports-up CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

This runs `scripts/deploy-cc-flink-reports.sh create`, which builds the PTF shadow JAR if missing, then runs `terraform apply -auto-approve` in `terraform/`.

- **First run takes ~6–8 minutes** (Kafka cluster provisioning dominates).
- **Re-applies are idempotent** — `CREATE ... IF NOT EXISTS` plus `lifecycle { ignore_changes = [compute_pool] }` on every statement.
- Optional fan-in provenance: `make cc-flink-reports-up ENABLE_MERGE_PROVENANCE=true ...` (script flag `--enable-merge-provenance=true`, default `false`). See [docs/flink-collector.md §2.4](#).
- Optional retention override: the script accepts `--day-count=<DAYS>` (default `30`); both `make` targets require `CONFLUENT_API_KEY` / `CONFLUENT_API_SECRET` or they fail fast with a clear message.

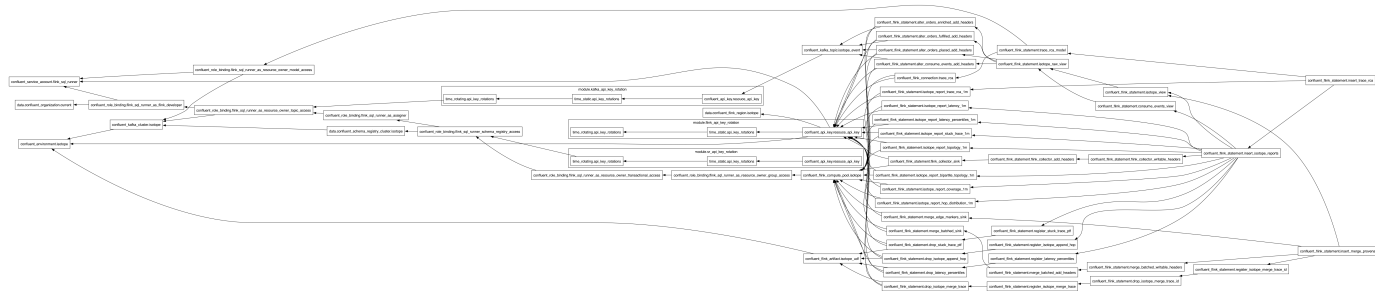
### 2.1 Verify in-band propagation

```
scripts/cc-app-run.sh place hello      # one traced record → orders.placed
scripts/cc-app-run.sh verify-inband    # asserts the Flink collector appended a hop
```

`verify-inband` reads a bounded snapshot of `orders.placed` and `orders.flink_enriched` and asserts that every trace on both topics kept its identity and gained **exactly one** hop, written by `flink-enrich`. Same trace ID with an incremented hop count is the proof that `ISOTOPE_APPEND_HOP` rehydrated the inbound isotope from the record's own headers rather than minting a fresh trace — the one propagation model that survives a Flink shuffle. It exits non-zero on failure, so it is usable as a CI gate. Pass a larger sample (`verify-inband 200`) if the two topics have drifted far apart.

The collector is a 1:1 projection with no window, so its output appears within seconds — the ~90s watermark wait applies only to the `TUMBLE` reports.

### 3.0 What gets created



| Resource                                                 | Name                                                                                                                                               | Notes                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>confluent_environment</code>                       | <code>confluent-kafka-isotope</code>                                                                                                               | ESSENTIALS stream-governance package                                                                                                                                                                                                                                                                                                              |
| <code>confluent_kafka_cluster</code>                     | <code>kafka-isotope</code>                                                                                                                         | Standard, single-zone, AWS us-east-1 by default                                                                                                                                                                                                                                                                                                   |
| <code>confluent_kafka_topic</code> × 4                   | <code>orders.</code><br>{ <code>placed</code> , <code>enriched</code> , <code>fulfilled</code> }<br>+<br><code>isotope_consume_edge_markers</code> | Only the event + consume-marker topics are pre-created. The 7 <code>isotope_report_*_1m</code> sink topics are created on first deploy by their <code>CREATE TABLE</code> (pre-creating them would make CCAF's Topic Catalog auto-import them as ( <code>key BYTES</code> , <code>val BYTES</code> ) and silently no-op the typed DDL).           |
| <code>confluent_service_account</code> + 6 role bindings | <code>isotope-flink-sql-runner</code>                                                                                                              | FlinkDeveloper (org) + ResourceOwner on topic/transactional-id/group/SR-subject * + Assigner                                                                                                                                                                                                                                                      |
| <code>confluent_flink_compute_pool</code>                | <code>isotope-flink-statement-runner</code>                                                                                                        | 10 CFU; headroom for the 8-INSERT statement set + ad-hoc SELECTs                                                                                                                                                                                                                                                                                  |
| <code>confluent_flink_artifact</code>                    | <code>isotope-flink-udf</code>                                                                                                                     | Uploads <code>ptf/build/libs/isotope-flink-udf.jar</code>                                                                                                                                                                                                                                                                                         |
| <code>confluent_flink_statement</code> × 28              | (see file)                                                                                                                                         | 6 ALTER TABLE + 3 VIEW + 8 sink CREATE TABLE + 5 CREATE FUNCTION (2 PTFs + 3 UDFs) + 1 EXECUTE STATEMENT SET holding all 8 INSERT INTOs (23 long-lived) + 5 transient DROP FUNCTION. The 2 merge UDFs are registered even when <code>var.enable_merge_provenance</code> is <code>false</code> — registration is inert until a statement calls it. |
| <code>confluent_flink_statement</code> × 3 (optional)    | (see <a href="#">terraform/setup-ccaf-ai.tf</a> )                                                                                                  | Optional AI trace-RCA report — <code>CREATE MODEL trace_rca</code> + 1 Protobuf sink + 1 <code>INSERT ... ML_PREDICT</code> . Gated on <code>var.enable_trace_rca</code> (default <code>false</code> ), so a normal apply skips them entirely. Set <code>rca_model_api_key</code> (and                                                            |

| Resource                                              | Name                                                            | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       |                                                                 | <code>rca_model_provider/_version/_endpoint</code> for a non-OpenAI provider) to enable. See root README §3.3.                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>confluent_flink_statement</code> × 5 (optional) | (see <a href="#">terraform/setup-ccaf-merge-provenance.tf</a> ) | Optional fan-in (merge) provenance — 2 sink <code>CREATE TABLE</code> + 2 header <code>ALTER TABLE</code> + 1 statement set holding the merge INSERT and its merge-edge INSERT. <b>Gated on</b> <code>var.enable_merge_provenance</code> (default <code>false</code> ). The 4 <code>CREATE/DROP FUNCTION</code> statements registering <code>ISOTOPE_MERGE_TRACE / ISOTOPE_MERGE_TRACE_ID</code> are <i>not</i> gated — registration is inert until a statement calls it. See <a href="#">docs/flink-collector.md §2.4</a> . |

Two rotating service-account API key pairs (one Kafka, one Schema Registry) are managed by `module.kafka_api_key_rotation` and `module.sr_api_key_rotation` in [terraform/setup-confluent-kafka.tf](#).

## 4.0 Useful outputs

```
terraform -chdir=terraform output environment_id
terraform -chdir=terraform output kafka_bootstrap_servers
terraform -chdir=terraform output schema_registry_url
terraform -chdir=terraform output -raw kafka_api_key      # sensitive
terraform -chdir=terraform output -raw kafka_api_secret   # sensitive
```

## 5.0 Drive traffic (required to see report rows)

The demo app runs **on your host** and talks to Confluent Cloud over SASL\_SSL. You don't export credentials manually: [scripts/cc-app-run.sh](#) sources [scripts/cc-cli-env.sh](#), which pulls the Kafka + Schema-Registry credentials from `terraform output`, builds the JAAS string, and invokes `./gradlew :app:run` with the six `-D` flags. It hard-fails if any of the seven required values is missing.

Run the 4-stage demo, one verb per terminal (same A/B/C/D order as the local demo):

```
scripts/cc-app-run.sh place 'hello'      # A — kick the chain off (orders.placed)
scripts/cc-app-run.sh enrich              # B — orders.placed → orders.enriched
scripts/cc-app-run.sh fulfill             # C — orders.enriched → orders.fulfilled
scripts/cc-app-run.sh ship                # D — terminal consume orders.fulfilled (emits
marker)
```

`cc-app-run.sh` also accepts the generic `send` / `hop` / `consume` / `sink` passthrough for ad-hoc topics — run it with no args for the full verb list.

**Sustained traffic.** The report jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, which only emit once the watermark advances past `window_end`. A burst inside a single 1-minute interval sits in one open window forever — spread records across **multiple** windows:

```
# 30 records, 5s apart ≈ 2.5 min of event-time → spans 3+ windows
for i in {1..30}; do scripts/cc-app-run.sh place "burst-$i"; sleep 5; done
```

Wait ~90s after the **last** record before checking results — that's the watermark catching up. `stuck_trace_alerts_1m` only fires for a trace that goes ≥60s of event time without a fresh hop; to exercise it, send one record to `orders.placed`, skip the `enrich/fulfill` hops, and keep sending unrelated records so the watermark crosses `event_time + 60s`.

## 6.0 Observe

In the Cloud Console **Flink SQL workspace**, query the report tables:

```
SELECT * FROM isotope_report_latency_1m;
SELECT * FROM isotope_report_bipartite_topology_1m;  -- all 6 edges: 3 produce + 3
consume
SELECT * FROM isotope_report_hop_distribution_1m;
-- ...coverage, stuck_trace, topology, latency_percentiles
```

Terminal D prints the same `trace_id` across all three hops. CCAF report topics ride **Protobuf+SR** (`proto-registry`), versus Avro+SR on the CP path.

## 7.0 Teardown

```
make cc-flink-reports-down CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

Runs `terraform destroy -auto-approve` — deletes **every** resource above, including the environment itself (the report sink topics are cleaned up via the environment cascade). Safe to run repeatedly.

**Cost note:** the Kafka cluster and compute pool bill while they exist. Teardown when you're done rather than leaving the environment running.

## 8.0 Troubleshooting

- **CONFLUENT\_API\_KEY ... must be set.** The `make` target requires both vars on the command line (or exported and passed through) — see §1.0 Prerequisites.
- **Aggregate functions are not supported.** Expected if you try to add a UDAF — CCAF rejects user-defined aggregates, which is why percentiles is a PTF — see root README §3.3.
- **Typed CREATE TABLE silently no-ops.** A report sink topic was pre-created, so the Topic Catalog auto-imported it as (`key BYTES, val BYTES`) — don't pre-create the `isotope_report_*_1m` topics — see §3.0 What gets created.
- **No report rows.** Almost always the watermark — traffic must span multiple 1-minute windows; wait ~90s after the last record — see §5.0 Drive traffic.
- **Column headers already exists in the table.** The four `ALTER TABLE ... ADD (headers ...)` statements are not idempotent and CCAF has no per-column `IF NOT EXISTS`; the column belongs to the topic's catalog entry and outlives the statement resource, so a re-created statement fails and leaves the resource tainted, blocking everything downstream. `make cc-flink-reports-up` now untaints them automatically before applying. Running terraform by hand? Do it yourself first: `terraform state list | grep _add_headers$ | xargs -n1 terraform untaint`.
- **error reading Flink Statement ... 401 Unauthorized.** Terraform reads a statement back with the API key recorded in `state`, and `module.flink_api_key_rotation` keeps only `number_of_api_keys_to_retain` keys — so a statement that outlives that many rotations points at a deleted key, and refresh fails on **both** apply and destroy (the environment becomes un-destroyable). `make cc-flink-reports-up/-down` now runs `scripts/tf-repair-flink-credentials.py` first, which re-points those statements at a key that still exists. Run it by hand if you drive terraform yourself: `python3 scripts/tf-repair-flink-credentials.py terraform/terraform.tfstate`.
- **Request is malformed. name needs to contain lowercase alphanumeric characters and hyphens only.** A `statement_name` used underscores. Statement names are Kubernetes-style (`isotope-report-latency-1m`); the sink `table` names keep their underscores (`isotope_report_latency_1m`) — the two are not the same string.
- **"stopped" or "properties\_sensitive" attribute must be updated for Flink Statement.** A statement resource changed in a way the provider cannot apply in place — edited SQL, a new `statement_name`, different

properties. Terraform plans it as an in-place update and the API rejects it; it has to be a destroy-and-recreate. `make cc-flink-reports-up` detects these from the plan and re-runs them as `-replace=` automatically; by hand, pass `-replace=confluent_flink_statement.<name>` to `terraform apply`.

- **App can't authenticate.** `cc-cli-env.sh` couldn't read `terraform output` — re-run `make cc-flink-reports-up` so the rotated keys exist, then retry.